03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 9

(JCD 6.1 - 6.2)

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Previous lecture

- Opacity can be defined as

$$\kappa_\nu = \sigma_R \frac{n}{\rho}$$

- Sources of opacity: electron scattering, free-free absorption, bound-free absorption, bound-bound absorption
- Energy equation:

$$\frac{dL}{dr} = 4\pi\, r^2 \left[\rho\epsilon - \rho \frac{d}{dt}\left(\frac{u}{\rho}\right) + \frac{P}{\rho}\frac{d\rho}{dt} \right]$$

Slido - Q&A only

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Outline

Energy transport by convection

Instability condition

Summary

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Energy transport by convection

- Remember the equation of radiative transport:

$$\boxed{\frac{dT}{dr} = -\frac{3\,\kappa\rho\,L(r)}{16\pi\,a\,c\,r^2\,T^3}}$$

- When the absolute temperature gradient is high, $\left|\frac{dT}{dr}\right| >> 0$, fluids become unstable and boil
- Examples of such instability regions: higher density layer on top of lower density layer (rapid temperature decrease leads to density increase...)
- Energy is then transported by convection - cyclic macroscopic mass motion: hot light elements rise and cool dense elements sink

Chemical element mixing

- Convection transports energy
- Convection also mixes the elements in a gas due to the motion of actual mass
- This chemical element "transport" in convective regions plays an important role in stellar evolution
- In many stars this is modelled adequately by the "mixing length" theory

Solar convective surface - cell size of Texas

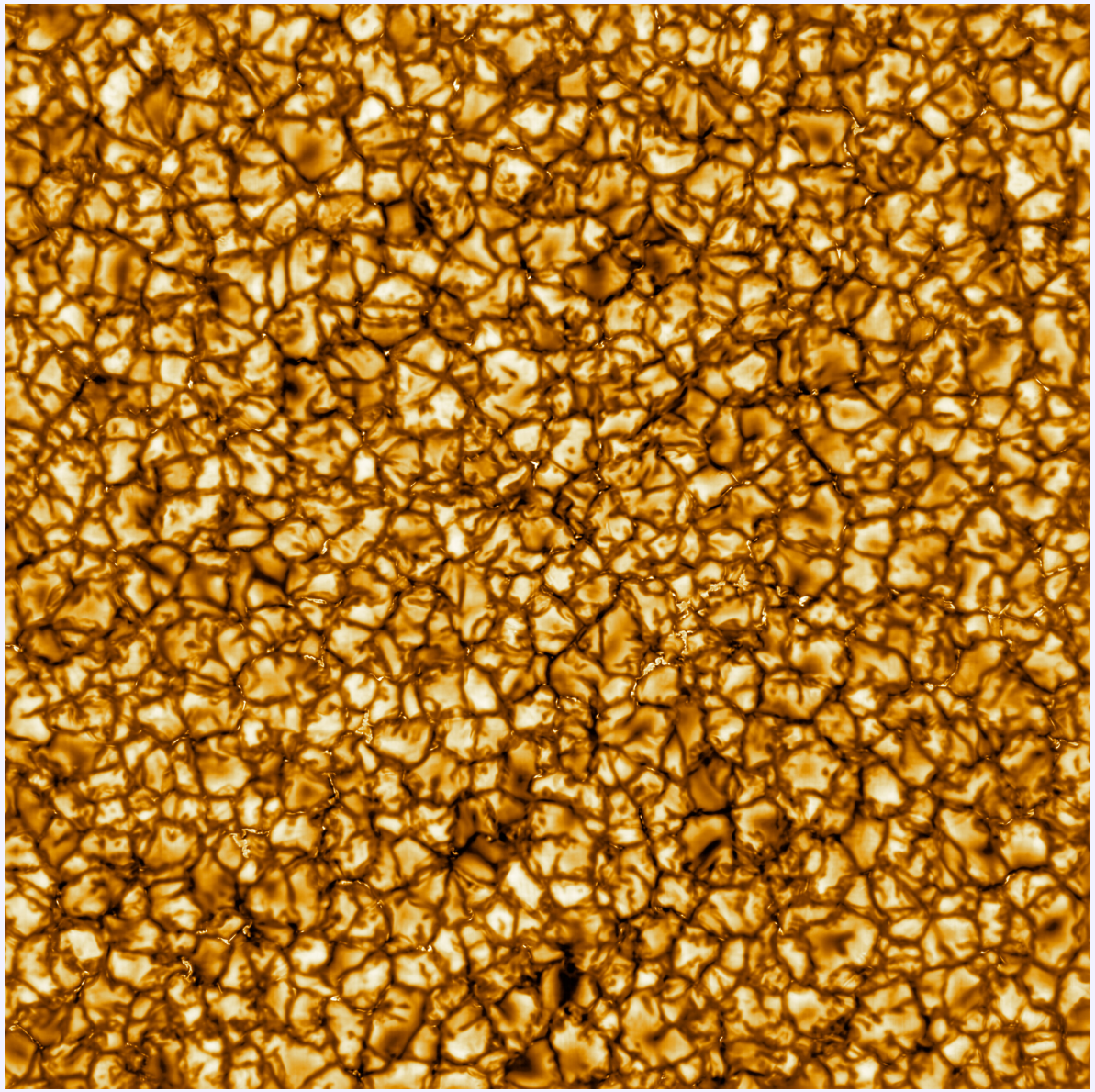
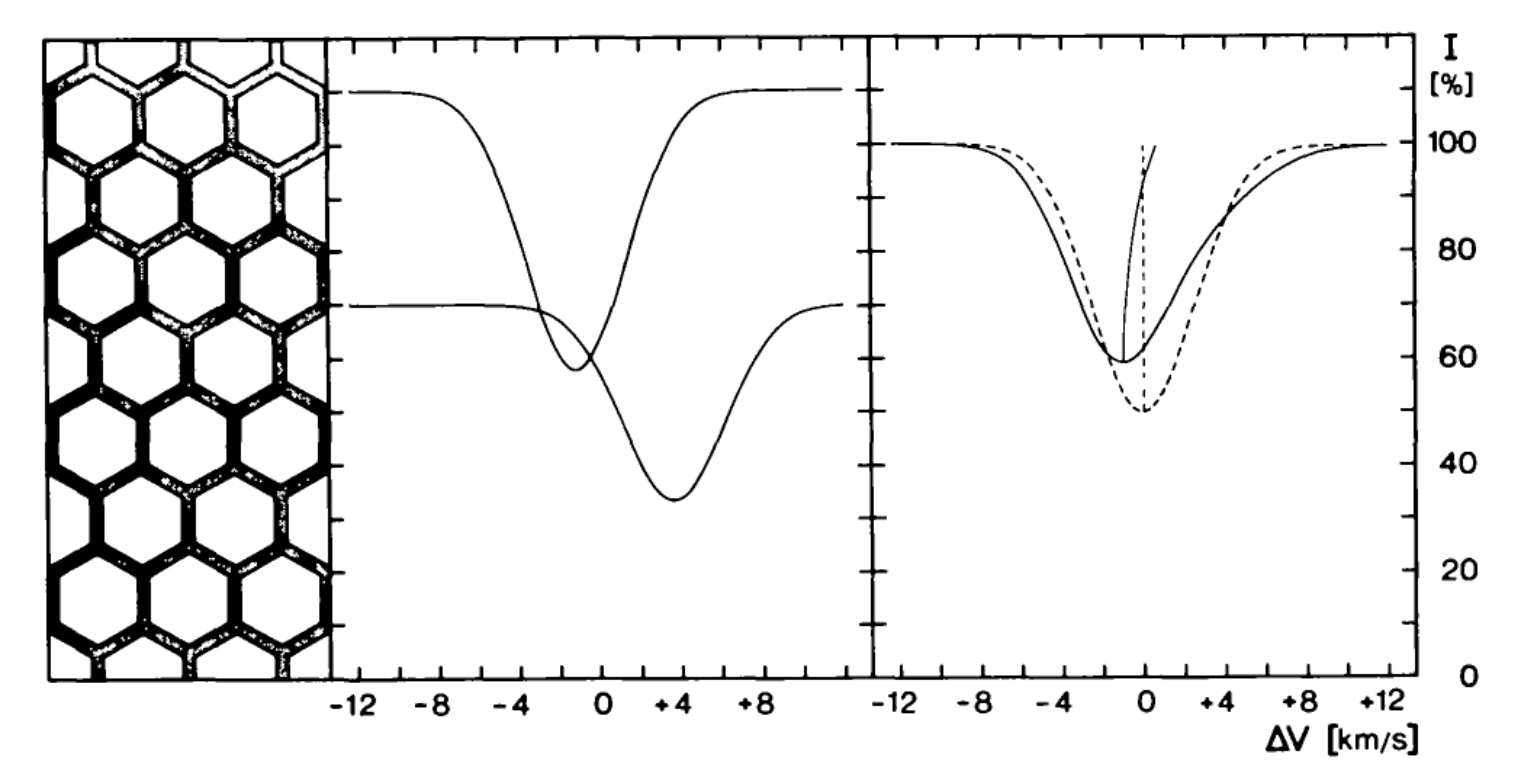


Image credit: NSO/NSF/AURA

Convection impact on spectral lines



Dravins et al. 1981

Stars with convective outer layers will have asymmetric and blueshifted spectral lines

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Situation

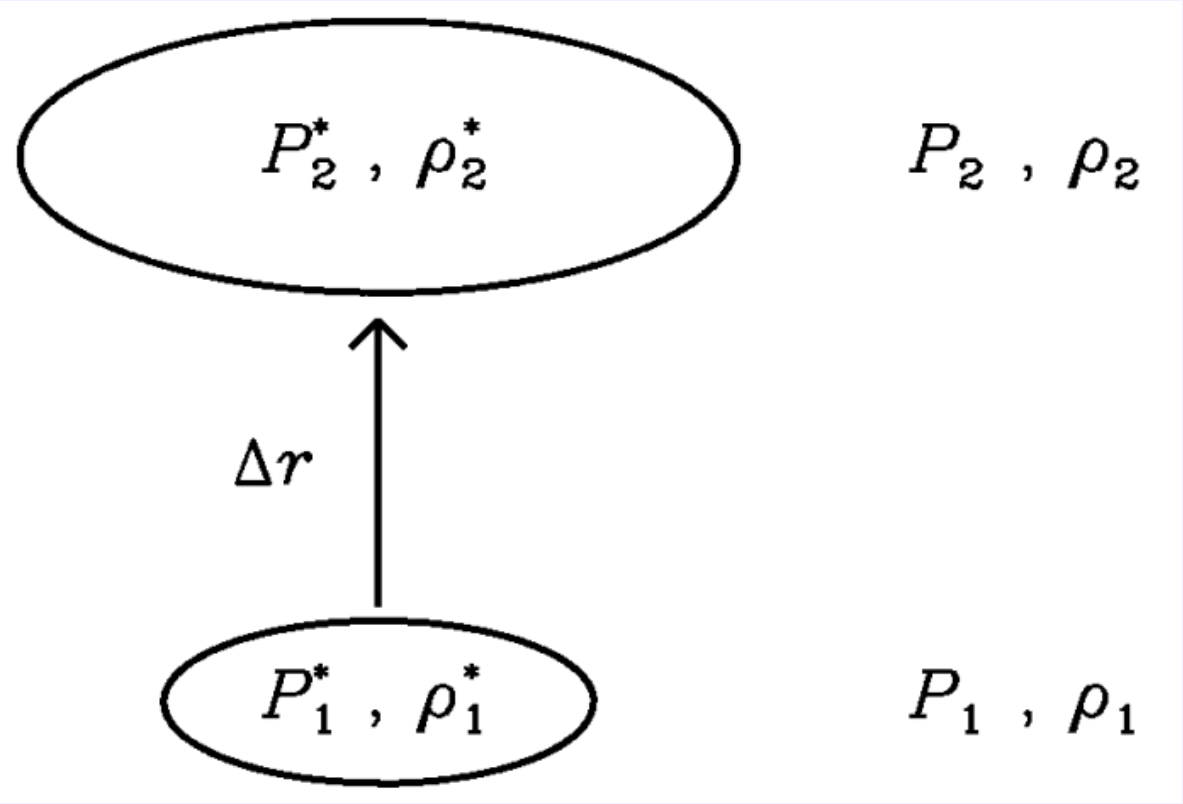

Image credit: JCD

- Element of gas moves distance Δr outwards
- The gas element as well as the outside region are described by their respective pressure and density
- Initially $P_1 = P_1^*$ and $\rho_1 = \rho_1^*$

Gas cell motion

- Motion is determined by buoyancy (think cork in water)
- At point 2, it holds that the force per unit volume is

$$f_\mathrm{b} = -g\left(\rho_2^* - \rho_2\right) = -g\,\Delta\rho$$

- For $f_\mathrm{b} > 0$ i.e. $\Delta\rho < 0$, the force is outwards and movement is accelerated: instability
- For $f_\mathrm{b} < 0$ i.e. $\Delta\rho > 0$, the force is inwards and gas cell has tendency to return: stability

Assumptions

- Motion is slow enough to have pressure balance between cell and its surroundings. Hence

$$P_2^* = P_2$$

- Motion is fast enough to occur adiabatically - there is no heat loss to the surroundings. Remember lecture 4:

$$\frac{d\rho^*}{\rho^*} = \frac{1}{\Gamma_1}\frac{dP^*}{P^*} = \frac{1}{\Gamma_1}\frac{dP}{P}$$

- Plausible assumptions since first assumption relates to dynamical timescale and second assumption to thermal timescale

Instability condition

- Need to find a criterion for $\Delta\rho$

$$\Delta\rho = \rho_2^* - \rho_2$$

- Remember the initial condition that $\rho_1 = \rho_1^*$
- Taylor expansion:

$$\rho_2 = \rho(r + \Delta r) \approx \rho(r) + \frac{d\rho}{dr}\Delta r = \rho_1 + \frac{d\rho}{dr}\Delta r$$

- We can show that

$$\Delta\rho = \left[\left(\frac{d\rho}{dr}\right)_{\text{ad}} - \frac{d\rho}{dr}\right]\Delta r$$

with

$$\left(\frac{d\rho}{dr}\right)_{\text{ad}} = \frac{1}{\Gamma_1}\frac{\rho}{P}\frac{dP}{dr}$$

Ledoux criterion

- Instability if $\Delta\rho < 0$. Hence

$$\frac{1}{\Gamma_1}\frac{\rho}{P}\frac{dP}{dr} = \left(\frac{d\rho}{dr}\right)_{\text{ad}} < \frac{d\rho}{dr}$$

- We can re-express this in terms of the logarithmic change:

$$\frac{d\ln\rho}{d\ln P} < \frac{1}{\Gamma_1}$$

where we used the result of hydrostatic equilibrium that pressure always decreases with increasing r.

Schwarzschild criterion - temperature gradient

- Remember the ideal gas law:

$$\rho = \frac{\mu m_u P}{k\,T}$$

- Asssume a constant mean molecular weight and fully ionised gas. We can show that

$$\frac{1}{\rho}\frac{d\rho}{dr} = \frac{1}{P}\frac{dP}{dr} - \frac{1}{T}\frac{dT}{dr}$$

- We can rewrite the convective instability criterion (lecture 4!) as

$$\frac{dT}{dr} < \left(\frac{dT}{dr}\right)_{\text{ad}} = \frac{\Gamma_2 - 1}{\Gamma_2}\frac{T}{P}\frac{dP}{dr}$$

- We can re-express this in terms of the logarithmic change:

$$\frac{d\ln T}{d\ln P} > \frac{\Gamma_2 - 1}{\Gamma_2}$$

where we again used the result of hydrostatic equilibrium that pressure always decreases with increasing r.

Ledoux versus Schwarzschild criterion

- Ledoux:

$$\frac{d \ln \rho}{d \ln P} < \frac{1}{\Gamma_1}$$

- Schwarzschild:

$$\nabla = \frac{d \ln T}{d \ln P} > \frac{\Gamma_2 - 1}{\Gamma_2} = \nabla_{\mathrm{ad}}$$

- Ledoux made no assumptions on chemical composition or ideal gas and is valid in regions where nuclear burning happens in contrast with the Schwarzschild criterion.

Summary

- Convection happens when absolute temperature gradient is high
- Ledoux criterion:

$$\frac{d \ln \rho}{d \ln P} < \frac{1}{\Gamma_1}$$

- Schwarzschild criterion:

$$\nabla = \frac{d \ln T}{d \ln P} > \frac{\Gamma_2 - 1}{\Gamma_2} = \nabla_{\mathrm{ad}}$$

NEXT LECTURE: Convective regions, mass-luminosity relations